# Forcing + Evidence — v0.3 UPDATE (2026-08-16): the operation dig sharpens Part I's forcing to a four-posit floor; rank=2 becomes the FIRST external forcing; Grace's boundary theorem explains WHY the dimension is a datum. Tier corrections folded in.

Keeper (audit); to be woven into the v0.2 flagship by Lyra. Source arc: K1593–K1624.

## Forcing + Evidence v0.3 UPDATE — the 2026-08-16 forced-object session

*Integrates into the v0.2 flagship (Lyra weave). Each item maps to a flagship Part; the flagship's honesty standard is unchanged — every line tiered, every correction kept in the text.*

### Headline: the forcing floor is now a NAMED FOUR-POSIT SET, and one integer is forced from OUTSIDE the geometry.

The flagship's Part I frames the claim as "forced modulo one datum (the generation count)." The operation dig (K1593–1610) makes the *mechanism* of that forcing explicit and, for the first time, forces one integer externally (non-circularly, without any physics datum). It also names the exact floor — four posits — where forcing stops and selection begins.

### Maps to Part I.1 (commitment as the stage-1 filter) — SHARPENED to a derivation

The flagship already says "commitment is the stage-1 filter" (E8 has no Hermitian symmetric space → no Bergman kernel → cannot host commitment). This session makes the filter a chain, all theorem/verified:

> binary commitment ⍰ rank 2 ⍰ isotropic complement ⍰ spin factor ⍰ type-IV (Lorentzian).

So "commitment requires a Hermitian symmetric structure" upgrades to "commitment (binary) *forces the type* to be type-IV" — the Lorentzian arena falls out of the act. And by the same isotropy, spacetime has no preferred frame (one principle, two jobs). [Tier: type forced mod the posits below; the isotropy = the T2565 spacetime keystone.]

### Maps to Part I.2 (forced modulo one datum) — the HONEST FOUR-POSIT ACCOUNTING

The v0.2 "forced modulo one datum" is correct but under-counted the floor. The honest set (nothing hidden):

- P0 — the arena: the commitment structure is a Euclidean Jordan algebra. Not a theorem; the Jordan–von Neumann–Wigner axiom.
- P1 — the act: a commitment is one *binary* distinction. An axiom ("one act is one act"), the cheapest possible; grounded by the Born rule as the resolving act (but Born is downstream, so it *grounds* not *forces*). ⍰ rank 2 ⍰ type-IV.
- P2a — exclusivity: "genuinely type-IV, no second name." Currently unmotivated; it does the dimensional work.

- P2b — minimality: and minimality is degenerate in EVERY lane (this session, Cal): it gives Jordan rank 1, the smallest single-error code [3,1,3], and domain n=3 — never the target. So n_C = 5 is NOT commitment-forced. It reduces to one gated equation (g = $N\_c^2$−rank, a rank-2 coincidence in a dense integer ring — possible-numerology, per the fishing count).

Net (sharper than v0.2): the TYPE is forced (mod P0+P1, two cheap axioms); the DIMENSION is the residual (P2a unmotivated + minimality degenerate). This *replaces* the vaguer “modulo one datum” with a precise floor — and it is *stronger* honesty, not weaker: the residual is named to the posit.

## ★ NEW — rank = 2 is forced from OUTSIDE the geometry (the first external forcing)

From Shannon (information theory, target-innocent, non-circular): a commitment’s binary alphabet = 2 = the Jordan rank. This is the first time any BST integer has been forced from outside the geometry — not from the physics we want to recover, not from the domain’s own invariants (which, by Grace’s theorem below, cannot force themselves). The information face doubles as the paper’s own title: Forcing + Evidence = 2 + 3 — rank 2 (the binary *decision* = the Forcing) + N_c=3 (the *validation*/parity = the Evidence). [Tier: rank=2 externally forced, Derived. N_c=3 = candidate — the coding d_min=3 → color-3 map is owed and obstructed (antisymmetry is d=2, detection not correction).]

## ★ NEW — Grace’s boundary theorem: WHY the dimension must be a datum (validates the whole frame)

A geometry cannot force its own defining parameters. Every invariant (eigenvalue, multiplicity, kernel dimension, curvature, the Born metric) is a *function of* the structure’s rank and dimension — so “invariant = 5 ⍰ n_C = 5” evaluates the function backwards (derivation-form, tautology-content). A theorem, general. This is the rigorous justification for the paper’s “one datum in” thesis: the five integers are the *definition-level boundary* of the theory — internal computation forces everything *given* them (the ~150 constants, every C/D theorem) but *cannot* force them. This is Gödel = boundary = definition (Casey’s Principle), made operational. The frame is not a hedge; it is a proven demarcation of where forcing stops.

## Maps to Part II (evidence) — TIER CORRECTIONS to fold in (this session)

- Koide → IDENTIFIED (was Conditional-Forced). The forcing was ruled BREAK on 2026-08-06 (K1221): the addresses {5/2,3/2,0} can never be $B_2$-symmetric, and the c-function magnitude route was retracted. The *relation* Q=2/3 stays banked (exact to 0.001%); the *derivation* does not. The honest geometric statement is “$A^2$=2 from equal-norm,” NEVER “$A^2$=rank” (a rank-2 coincidence). Koide is open in physics — the field has not derived it either. Falsifier: $m_\tau$=1776.969 MeV (0.91$\sigma$, dies at $\sigma\approx0.03$) — attribute to Koide 1981, not BST (BST inherits only if a derivation returns 2/3; none does).
- ★ DO NOT CITE the retracted lepton nested-tower / $c_5/c_3=\Gamma(5)/\pi^2$ (F669, K853, K928, K1221 — retracted 6+ times, a named repeat offender). The $\nu$-addresses {5/2,3/2,0} (T2517) ARE banked; the *genus-{5,3,0} nested sub-domains* are NOT — same shape, different claim.
- The operation’s ENCODE face is “self-validating,” NOT “unforgeable” (a public code forges trivially; tamper-resistance is the irreversibility). It is a d=2 detection/erasure code, not error-correction. Anywhere the paper describes the confined-3 as “error-correcting” or the record as “unforgeable,” correct it.
- QM stays DONE (10/10 axioms, zero posits) — unchanged, the standout; now stated as “measurement = the operation” (the commitment ontology is the Internal-C artifact, drafted this session: Paper_The_Operation_One_Page_v0.1).

## Maps to Part 0.5 (the one-page) — a companion now exists

The self-contained operation synthesis is drafted: Paper_The_Operation_One_Page_v0.1_2026-08-16.md (one geometry, one act [WRITE+ENCODE], three readings, the four-posit floor, the honest tiers). Part 0.5 and that one-page should be reconciled to one voice (Lyra) — the one-page is the “one act” framing; Part 0.5 is the “five operations” framing; they are the same object read two ways (state that, count once).

## The net for the paper’s thesis

The v0.2 thesis — “forced modulo one datum, GR-level in input economy” — is sharpened and strengthened: the forcing floor is now a named four-posit set; the *type* is forced by a two-axiom chain (with rank=2 forced *externally* for the first time); and Grace’s boundary theorem *proves* why the remaining integers must be data, not derivations. The honesty is the argument, and the argument is now more precise: we can say exactly which two axioms carry the type, which one integer we forced from outside, and why the rest are the definitional boundary.

— Keeper, v0.3 update, 2026-08-16. Weave into the flagship (Lyra). Tiers current; corrections in-text by design. Nothing pushed. CP existence-only.